Mahadesh Chandro Mondal

Email: mahadesh.mondal9504@gmail.com | **Contact:** +880-1976-608538 | **Research Portfolio:** https://mahadeshmondal9504.github.io/
Linked in: https://www.linkedin.com/in/mahadesh-mondal-854442432/ | **ORCID:** 0009-0006-2742-9714
Google Scholar: https://scholar.google.com/citations?user=jkIjqokAAAAJ&hl=en&authuser=1

EDUCATION

University of Dhaka *B.Sc. in Applied Chemistry & Chemical Engineering*	**CGPA 3.47/4.00 | 2022-2026**
Govt. Sundarban Adarsha College, Khulna *Higher Secondary Certificate (HSC)*	**GPA 5.00/5.00 | 2018-2020**
Khulna Textile Mills High School *Secondary School Certificate (SSC)*	**GPA 5.00/5.00 | 2016-2018**

RESEARCH PROFILE

Interdisciplinary chemistry researcher with a background in Applied Chemistry and Chemical Engineering at the University of Dhaka, with a clear vision to contribute to the fields of pharmaceutical chemistry, drug delivery, nanomedicine, computational chemistry, and advanced functional materials. My research experience includes studying drug loading and release mechanisms for sustained drug delivery, protein- and biopolymer-based drug carriers, functionalized biomaterials for targeted therapeutics, nanoparticles, and nanocomposites. This interdisciplinary experience has further motivated my interest in molecular modeling and computational & data-driven research, particularly in designing functionalized flavonoid derivatives for cancer therapeutics. I also have research experience in advanced functional materials and environmental applications, including graphene-based nanocomposites, alloys, and nanomaterials for applications in oil recovery, photocatalysis, and biomedicals. I am particularly interested in integrating medicinal chemistry, organic synthesis, and nanotechnology with computational chemistry and molecular modeling to develop innovative drug delivery platforms and therapeutic strategies. My long-term research goal is to bridge experimental and computational approaches in drug discovery, pharmaceutical development, and nanomedicine.

RESEARCH INTERESTS

Drug Delivery, Drug Loading & Release Mechanism, Computer-Aided Drug Design, Molecular Docking & Molecular Dynamics, Machine Learning & AI-Assisted Drug Discovery, Protein-Ligand Interaction & Protein Biochemistry, Cancer Studies & Therapeutics, Organic Synthesis, Pharmaceutical & Medicinal Chemistry, Nanomedicine & Nanoparticles.

SKILLS

In-Process skills:	Centrifugation, Filtration, Ultrasonication, Lyophilization, Dialysis, Reverse Osmosis, Membrane Casting, Homogenization, Distillation, Sterilization, Evaporation, Extraction, Neutralization.
Analytical & Material Characterization:	Molecular Docking, Dynamics Simulation, MM/GBSA Analysis, XRD, FTIR, UV Spectroscopy, SEM, TGA, Gas Chromatography, HPLC, PID.
Simulation & Software Skills:	PyMOL, AutoDock Vina, RDKit 3D, Mol* Viewer, OpenMM, AmberTools, VS Code, Aspen HYSYS, ChemDraw, ImageJ, OriginPro, Adobe Photoshop, Krita, Microsoft Office, LaTeX.
Operational Skills:	Heat Exchanger, Centrifugal Pump, Multiple Effect Evaporator.
Organizational Skills:	Event & Conference Management, Project Coordination & Distribution, Strategic Planning, Grant Acquisition, Team Leadership, Anchoring, Volunteer and Stakeholder Management, Research and Administrative Responsibilities.
Language & Communication Skills:	Bangla (native), English: IELTS Band Score: 7 (L: 7.5, R: 7.0, W: 6.5, S: 6.0).

RESEARCH ASSISTANTSHIP

Undergraduate Research Assistant | *Professor (Grade-I) Dr. A.M. Sarwaruddin Chowdhury* | Applied Chemistry & Chemical Engineering, University of Dhaka.	**2024-2026**

- Conducted sustained research activities in computer-aided drug design, molecular docking and dynamics simulation, drug delivery systems, nanomaterials, alloys, graphene-based composites, adsorption, and environmental remediation.
- Contributed to literature review, scientific writing, and preparation of review manuscripts, including two published and two under-review articles. Also, currently contributing to ongoing research on sustainable materials and waste valorization.
- Assisted in experimental research, material characterization, laboratory management, commercial product sample testing, research data analysis, and related administrative responsibilities.

PUBLICATION & PROJECT INFO

Published Articles

1. ***Mahadesh Chandro Mondal***, H.M. Wasi Uddin, Sania Ashrafi, *et al.* "**Encapsulation of Protein-Based Drug Carriers for Controlled Loading and Releasing Efficiency: A Review.**" *The Protein Journal, 2026, doi: https://doi.org/10.1007/s10930-026-10346-4*
2. ***Mahadesh Chandro Mondal***, H.M. Wasi Uddin, Qazi Riti Mahpara Progoti, *et al.* "**Enhanced oil recovery via metal oxide nanoparticles: A review of recent advancements, core mechanisms, and future outlook.**" *Next Materials, 12, 102004 (2026), doi: https://doi.org/10.1016/j.nxmate.2026.102004*
3. H. M. Wasi Uddin, Qazi Riti Mahpara Progoti, ***Mahadesh Chandro Mondal***, *et al.* "**Adaptive nickel–titanium shape memory alloy for smart systems: Mechanisms, manufacturing, and applications across biomedical, aerospace, civil, and energy.**" *Materials Today Communications, 50, 114564 (2026), doi: https://doi.org/10.1016/j.mtcomm.2025.114564*
4. Tasfik Saif, H.M. Wasi Uddin, Qazi Riti Mahpara Progoti, ***Mahadesh Chandro Mondal***, *et al.* "**Comparative Gamma and Neutron Shielding Performance of High-Entropy and Shape Memory Alloys: A Data-Driven Analysis.**" *Nuclear Analysis, 100279 (2026), doi: https://doi.org/10.1016/j.nucana.2026.100279*
5. H.M. Wasi Uddin, Qazi Riti Mahpara Progoti, ***Mahadesh Chandro Mondal***, *et al.* "**Shape Memory Alloys in Biomedical Engineering: A Scopus-Based Bibliometric Analysis and Research Trajectory Forecasting via Machine Learning.**" *Biomedical Materials & Devices, 2026, doi: https://doi.org/10.1007/s44174-026-00796-w*

6. Qazi Riti Mahpara Progoti, H.M. Wasi Uddin, ***Mahadesh Chandro Mondal***, *et al.* "**FTIR-Based Quantification of Microplastics: Scopus-Based Bibliometric Analysis and Forecasting of Research Trajectories with Artificial Intelligence (AI).**" *Journal of Hazardous Materials: Plastics, 5, 100079 (2026), doi: https://doi.org/10.1016/j.hazmp.2026.100079*
7. Golap Roy, ***Mahadesh Chandro Mondal***, *et al.* "**Chapter 7 - Electronic structure and photocatalytic activity of BiOX photocatalyst.**" *Book: BiOX-based Photocatalysts for Dual Applications (Elsevier), doi: https://doi.org/10.1016/B978-0-443-23861-1.00007-9*

Manuscripts under Review

1. ***Mahadesh Chandro Mondal***, Md. Jabbir Ali, *et al.* "**Computational Design and Evaluation of Chr-CH_2-Naphthyl Derivative Targeting the MCL-1 BH_3-Binding Groove via Molecular Docking and Dynamic Simulations for Apoptosis-Based Cancer Therapy.**" *Pharmacological Research - Reports (Manuscript Under Review), Manuscript Number: PREREP-D-26-00437.*
2. ***Mahadesh Chandro Mondal***, Arpita Dutta, *et al.* "**Functionalization and Ligand Conjugation of Novel Chitosan Nanoparticles for Targeted Drug Delivery System: A Review.**" *Manuscript Under Review*.
3. H.M. Wasi Uddin, Qazi Riti Mahpara Progoti, ***Mahadesh Chandro Mondal***, *et al.* "**Shape Memory Alloys in Aerospace Engineering: A Bibliometric and Machine Learning Analysis of Research Trends and Future Directions.**" *Advanced Light Materials (Manuscript Under Review), Manuscript Number: ALMATE-D-26-00032.*

Projects

1. **Simulation of BTX Recovery from Petrochemical Wastewater Using Aspen HYSYS.**
   Simulated the recovery of benzene, toluene, and xylene (BTX) from petrochemical wastewater using Aspen HYSYS, incorporating process separation and optimization strategies. Evaluated process performance and product recovery through steady-state process simulation.
2. **Industrial Production of Cellulose Nanocrystals from Jute Fiber: Process Design, Optimization, and Economic Assessment in Bangladesh.**
   Designed and evaluated an industrial-scale process for producing cellulose nanocrystals (CNCs) from jute fiber, incorporating process optimization, mass and energy balances, and economic assessment for a 500-tonne/year plant.
3. **Synthesis and Development of a Sustainable Graphene Oxide-Fe_3O_4-Cellulose Acetate Nanocomposite for Efficient Heavy Metal Adsorption from Wastewater.**
   Contributed to the synthesis of graphene oxide (GO) and fabrication of GO-Fe_3O_4-Cellulose Acetate nanocomposite. Also contributed to some analytical and material characterizations, including FTIR, SEM, and TGA.

Selected Ongoing Research

- ***Mahadesh Chandro Mondal***, A.M. Sarwaruddin Chowdhury, *et al*. "**Sustainable Chicken Feather-Derived Waste Natural Fiber-Reinforced Reduced Graphene Oxide (rGO)/g-C_3N_4 Ternary Composite for Environmental Assay & Visible-Light-Driven Photocatalytic Degradation of Organic Dyes.**"

Extracted natural fibers from chicken feather waste and fabricated reduced graphene oxide (rGO) to develop a sustainable ternary composite. Conducted contact angle measurements, water/oil uptake analyses, and degradation studies in acidic and basic environments for environmental assays. Also conducted FTIR, TGA, and tensile strength analyses to evaluate the composite's surface, thermal, and mechanical properties. Incorporation of g-C_3N_4 and visible-light-driven photocatalytic degradation studies are ongoing.

ACHIEVMENTS & AWARDS

- Recognized as one of the **Top 20 Most Prolific Authors** of the Department of **Applied Chemistry and Chemical Engineering, University of Dhaka**, in **Scopus-Indexed Literature (AF-ID 60014714)** of articles and reviews published in 2026 (Jan 1 - Sep 10, 2026).
- **Champion** at **BUET CHE Video Competition, 2024**
  Topic: "Chemical Safety: Think Safe, Work Safe."
- **Champion** at **ACSDU Poster Presentation Competition, 2023**
  Topic: "Fueling the Future: Artificial Photosynthesis"
- **Textile Hack 2024:** Selected as one of the top 10 candidates from approximately 300 applicants.
  Topic: "Metal Meets Dye: Integrated Solutions for Textile Wastewater Treatment"

INDUSTRIAL TRAINING PROGRAM

Certified trainee at Training Institute for Chemical Industries (TICI)	**Grade: A+ | January, 2026**

Completed a 4-week intensive industrial training program on Industrial Process Unit Operations and Process Control Techniques, covering process technology, industrial water treatment, pump and compressor operation, heat exchangers, boilers and steam systems, process Instrumentation and control (PLC, SCADA & PID), chromatography (GC/HPLC), FTIR, EDXRF, industrial safety, and process simulation, with extensive hands-on practical and demonstration-based training.

DEVELOPMENT & LEADERSHIP EXPERIENCE

Administrative Secretary at ACS International Student Chapter, University of Dhaka (ACSDU)	**2024-2025**

Coordinated national scientific outreach and environmental initiatives in Bangladesh to promote chemistry and sustainability. Organized key events, including the **1st International Conference on Environmental Sustainability & Green Earth (ICESGE-2025)** and the **World Environment Day Poster Competition**; scientific outreach, including **ChemQuest-2024** and the **Coastal Chemistry Carnival 2025**, as well as various environmental campaigns and webinars. Contributed to strategic planning and secured international grants totaling USD 5,500 (2025 Festival Grant: USD 3,000; 2025 IAC Global Innovation Grant: USD 2,500).

General Secretary at We for Us Foundation BD	**2019-Present**

Provided leadership for a non-governmental organization focused on education, humanitarian assistance, and livelihood development in underprivileged communities in Bangladesh. Key initiatives included "**Education for All**", "**Project Shabolombi**", "**Project Eid Happiness**", and "**Warming the Souls**". Coordinated COVID-19 relief efforts, facilitating donations of approximately 50 oxygen cylinders to hospitals and economically disadvantaged patients during the 2020-2021 pandemic.

Volunteer at 13th Bangladesh Chemistry Olympiad	**May, 2024**

Volunteered and provided the necessary assistance to the participants in the Preliminary Round (Online) at the Department of Applied Chemistry & Chemical Engineering, University of Dhaka on 3rd May, 2024, and in the Final Round (Offline) at the Curzon Hall on 17th May, 2024.